Sajeeb Kumar Ray

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Overview

Dedicated to building a career in academia with a strong focus on teaching, research, and student mentorship. My objective is to advance cutting-edge research in artificial intelligence and emerging technologies, while fostering an engaging learning environment that cultivates problem-solving, innovation, and analytical thinking.

Education

MS (Ongoing) **Master's by Research** in Department of Computer Science and Engineering, University of Yamanashi, Japan (Japanese Government MEXT Scholarship Awardee).

B.Sc. (2023)
(Held in 2025) **Bachelor of Science in Information and Communication Engineering** Department of ICE, Pabna University of Science and Technology, Bangladesh. **CGPA: 3.91/4.00 (First Position; Honours) [Dean's List] [Prime Minister's Gold Medal 2025 Nominee]**.

HSC (2019) **Higher Secondary Certificate**, Collectorate School and College, Rangpur, Bangladesh. Group: Science, Board: Dinajpur. GPA: 5.00/5.00.

SSC (2017) **Secondary School Certificate**, Jaldhaka Pilot High School, Nilphamari, Bangladesh. Group: Science, Board: Dinajpur. GPA: 5.00/5.00.

Experience

Lecturer **Feb 2026 – Sep 2026 (Full Time)**
Department of Computer Science and Engineering, Varendra University

- Taught Digital Image Processing.
- Mentored students in the Varendra University Programming Club.
- Orgranized Code Harbor Season - 5 (Intra University Programming Contest)

Research Assistant May 2024 – Jun 2025 (Part Time)
Dept. of ICE, Pabna University of Science and Technology

- Developed a Dual-Head CNN (DH-CNN) for plant leaf classification and disease detection. Achieved 99.71% accuracy in classification and 99.26% in disease identification, and a Multi-Task CNN (MTL-CNN) with ResNet-50 for freshness detection of fruits and vegetables, achieving 98.63% accuracy.
- Jointly developed RViT-FusionNet, a hybrid CNN–Transformer model with Local Cross-Attention for brain tumor classification via MRI, achieved 99.66% accuracy Grad-CAM interpretability.
- Built XR-U-Net for lung image segmentation, achieving 95.7% accuracy on chest X-rays.
- Constructed a CNN-RNN framework for Bangla image captioning using EfficientNetB4 + ResNet-50, achieving BLEU score of 0.54.

- Jointly developed YOLOv8-based vehicle detection system tailored for Bangladeshi highways and a CNN model for traffic sign recognition, achieving 99.68% accuracy on GTSRB dataset.
- Jointly built a microcontroller + RTC-powered AC device controller for scheduled automation and an IoT-based Smart Product Price Display System for the Bangladesh retail sector.

Independent Researcher

- Evaluated classical, learning-based, and foundation models for 3D ultrasound image registration.
- Developed a dual-encoder dual-classifier ViT–CNN model with Local Cross-Attention for leaf analysis, achieving 99.98% accuracy in 30-species identification and 99.46% accuracy in 125-disease detection.
- Designed a multi-layer perceptron (MLP) model for Human Activity Recognition (HAR) using UCI-HAR dataset, reaching 97% accuracy.

Experienced in working with large-scale datasets (200,000+ images and 160,000+ Bengali sentences), evaluating foundation models, and designing state-of-the-art system architectures.

Publications

Journal:

1. N. Islam[❶], **S. K. Ray**[❶], M. A. Hossain, and S. Islam, "RViT-FusionNet: A Local Cross-Attention Feature Fusion-based Hybrid Framework for Brain Tumor Classification," *Neural Computing and Applications*, 2026, 10.1007/s00521-026-12290-x [Springer-Q1] [Cite Score: 12.7] [27 Pages].

   ❶ These authors contributed equally to this work.

2. **S. K. Ray**, M. A. Hossain, N. Islam, and M. A.F.M. R. Hasan, "Enhanced plant health monitoring with dual head CNN for leaf classification and disease identification," *Journal of Agriculture and Food Research*, Vol. 21, 2025, 10.1016/j.jafr.2025.101930 [Elsevier-Q1] [Cite Score: 9.8] [Impact Factor: 7.2]

3. **S. K. Ray**, A. Kumar, N. Islam et al., "Freshness detection of fruits and vegetables using multi-task convolutional neural network and ResNet-50," *Bulletin of Electrical Engineering and Informatics*, 2026, [10.11591/eei.v15i4.11368] [Accepted] [Q3; Cite Score: 4.2]

4. N. Islam, **S. K. Ray**, M. Mynoddin et al., "Understanding the Traffic Sign through a Deep CNN Architecture for Enhanced Traffic Safety," *Annals of Emerging Technologies in Computing*, Vol. 13, pp. 429–438, 2026, 10.33166/AETiC.2026.02.003 [Q2; Cite Score: 4.0].

5. M. A. Hossain, **S. K. Ray**, N. Islam et al., "Enhanced human activity recognition through deep multilayer perceptron on the UCI-HAR dataset," *International Journal of Advances in Applied Sciences*, Vol. 13, pp. 429–438, 2024, 10.11591/ijaas.v13.i2.pp429-438 [Q4; Cite Score: 1.0]

Conferences:

1. A. Islam, S.A. Oishi, ...**S. K. Ray***, "GR-ACE Net: A Hybrid Graph-Attentional Framework with Global Relational Reasoning for Deepfake Forensics", *International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure*, Pabna, Bangladesh, 2026, 10.1109/PECCII70991.2026.11661873[Scopus Indexed]

2. A. Sharma, **S. K. Ray**, N. Islam et al., "An End-to-End Attention-Based Framework for Bangla Image Captioning", *2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking*, Chattogram, Bangladesh, 2026, 10.1109/QPAIN69676.2026.11546644[Scopus Indexed]

3. A. Islam, **S. K. Ray**, S.A. Oishi et al., "A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images", *2nd International Conference on Quantum*

Photonics, Artificial Intelligence & Networking, Chattogram, Bangladesh, 2026, 10.1109/QPAIN69676.2026.11545784 [Scopus Indexed]

4. S.S.M. Pranto, M.O. Faruk, Alamin, **S. K. Ray** et al., "Comparative Evaluation of CNN Backbones with Explainable AI for Multi-Class Chest X-Ray Classification", *2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking*, Chattogram, Bangladesh, 2026, 10.1109/QPAIN69676.2026.11546150[Scopus Indexed]
5. Alamin, T. Debnath, **S. K. Ray** et al., "IoT Based Smart Product Price List Display Board," *28th International Conference on Computer and Information Technology*, Cox's Bazar, Bangladesh, 2025, 10.1109/ICCIT68739.2025.11490213 [Scopus Indexed]
6. M. K. Saha, M. S. Hosain, M. R. Hossen, **S. K. Ray** et al., "Speech Emotion Recognition from Bone-Conducted Speech Using Wav2Vec2 Transformer Model," *7th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, Narayanganj, Bangladesh, 2025, 10.1109/STI69347.2025.11367517 [Scopus Indexed]
7. **S. K. Ray**, A. Islam, M.C. Chanda et al., "Deep Learning Based Lung Image Segmentation Using XR-U-Net," In *27th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2024, 10.1109/ICCIT64611.2024.11022346 [Scopus Indexed]
8. M. A. Hossain, M. A. R. Hasan, **S. K. Ray**, and N. Islam, "Generating Bangla Image Captions with Deep Learning Techniques," In *6th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, Narayanganj, Bangladesh, 2024, 10.1109/STI64222.2024.10951094 [Scopus Indexed]
9. N. Islam, **S. K. Ray**, M. A. Hossain et al., "Vehicle Classification and Detection Using YOLOv8: A Study on Highway Traffic Analysis," In *International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*, Rajshahi, Bangladesh, 2024, 10.1109/ICRPSET64863.2024.10955913 [Scopus Indexed]
10. Alamin, **S. K. Ray**, M. A. Hossain et al., "Embedded Systems-Based AC Device Controller," In *International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*, Rajshahi, Bangladesh, 2024, 10.1109/ICRPSET64863.2024.10955881 [Scopus Indexed]

Research, Programming & Computer Skills

Research	Data Analysis (Data Cleaning, Missing Value Handling, Outlier Detection, Feature Engineering, Data Preprocessing), Customized and Pretrained System Architectures (CNN, RNN, LSTM, ViT using PyTorch, TensorFlow), Visualisations
Competitive Programming (May 2020 – Mar 2023)	Top rating **4-star (1816) on CodeChef**; **Specialist (1492) on CodeForces**. Solved 750+ problems across Codeforces, HackerRank, Beecrowd. Problem Setter: Intra-University Programming Contest 2025.
Operating Systems	Windows, Linux.
Languages	Python, C, C++, R, MATLAB.
Others	Proteus, XAMPP, mikroC, PICkit, SQL Server, PCB Design, Circuit Design.

Embedded Systems & Robotics Projects

Jan 2024 - Mar 2024	**AI Baby Nursing Robot** – An automated baby rocker capable of classifying baby cry to understand baby's needs. It can provide some necessary information such as Gas leakage, Smoke detection, and Temperature, Humidity, and CO level of the home environment.
Aug 2023 - Oct 2023	**Infant Monitoring System** – A low-cost, mobile-based system was proposed to monitor infants in the workplace. The system uses IoT devices and real-time video transfer to monitor infants and detect safety hazards.

Jul 2023 - Aug 2023	**Hardware Trial Security Circuit** – Industrial hardware trial-run device with password-based security. Designed with PIC16F877A microcontroller. PCB and circuit design.
May 2023 - Jun 2023	**RC Plane** – Radio-controlled surveillance aircraft. Role: debugging errors and supporting teammates (with Alamin & Gulam Mustofa). Hardware: transmitter, receiver, servo motor, BLDC motor, etc.

Achievements

- **Dean's List** for achieving CGPA over 3.80 out of 4.00.
- **Student of the Year Award – 2025** for academic excellence, research contributions, innovation achievements, and leadership in multiple university organizations.
- **Finalist, Disaster Hackathon 2.0** – Designed a real-time cyclone alert system for coastal regions facing power and network outages. Secured a top 5 finalist position and refined the solution through intensive mentoring, with potential for real-world implementation.
- **Champion, Innovation Showcasing 2024** – Presented a cost-effective smart room concept. to assist physically challenged individuals, especially those from middle-income families.
- **Champion**, ICE Fiesta Programming Contest 2023 & **Runner-Up**, ICE Fiesta Programming Contest 2024.
- **University Merit Scholarships** (2020, 2021, 2022, and 2023).
- Competitive Programming Ranks: Codeforces Round 857 Div.2 (3rd in Bangladesh), CodeChef Starters 79 Div.2 (5th in Bangladesh), CodeChef Starters 49 Div.4 (4th in Bangladesh).

Talks & Presentations

Apr 2026	Keynote talk on "An Undergraduate's Guide to Starting Strong", delivered at the Career Session of 6th Freshers' Orientation Programme 2026, organized by Solver Green, Skill Development Club, PUST.
Mar 2025	Research talk on "Autonomous Plant Health Monitoring using DH-CNN" in the Workshop on "From Ideas to Impact: Collaborative Research Opportunities between PUST and the University of Yamanashi, Japan".
Feb 2025	Seminar on "Fresher's Guide to Research" – an introductory session designed for undergraduate students with no prior experience in research, organized by Solver Green, Skill Development Club, PUST.
Dec 2024	International Conference talks: (1) "Deep Learning Based Lung Image Segmentation Using XR-U-Net", ICCIT 2024, Cox's Bazar. (2) "Generating Bangla Image Captions with Deep Learning Techniques", STI 2024, Narayanganj. (3) "Vehicle Classification and Detection Using YOLOv8", ICRPSET 2024, Rajshahi.
Oct 2024	Finalist presentation as team lead at Disaster Hackathon - 2.0; Presented a solution to deliver real-time cyclone forecasts from BMD to remote coastal regions without electricity, internet, or cellular network access.
May 2024	Innovation Showcasing 2024 – Presented "Smart Bedroom Technology" concept aimed at supporting physically challenged individuals from middle-income families, in Innovation Showcasing-2024, PUST.

Certifications

- Advanced Learning Algorithms (Coursera).
- Supervised Machine Learning: Regression and Classification (Coursera).

- Problem Solving (Intermediate) – HackerRank.
- Problem Solving (Basic) – HackerRank.

Volunteering

- Vice President – ICE Association (Jan 2024 – Sep 2025). Led student governance by launching three departmental clubs, supporting research initiatives, securing scholarships, and organizing the 1st Alumni Reunion & Fiesta 2025 along with cultural and sports events.
- Vice President – Solver Green (Sep 2024 – Nov 2025). Led initiatives to expand research opportunities by organizing seminars, workshops, and launching new courses. Advocated for early student involvement in research, ensuring that junior students gained valuable experience alongside final-year students to foster a stronger research culture.
- Class Representative – Dept. of ICE, PUST (Feb 2022 – July 2025). Managed class schedules, facilitated teacher-student communication, and supported peer well-being.
- Founding Member & Former Joint Organizing Secretary – PUST Career & Entrepreneurship Club (Nov 2024 – Present). Bridging the industry-academia gap.

Languages

- Bangla — Native Language
- English — IELTS: 6.5 (L:7.5; R:6; W:6; S:6) and medium of instruction during B.Sc. in Engineering.

References

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I hereby declare that all the information provided here is authentic.

Signed
Date: 30.10.2026